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Control-Oriented Design and Performance Evaluation of a Three-Terminal Storage-Assisted EV Fast-Charging System with an Interleaved Bidirectional Buck–Boost Converter

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***Abstract*—This paper presents a three-terminal storage-assisted bidirectional EV fast-charging system based on a common DC-link that coordinates a grid-side three-phase active rectifier, an EV-side interleaved bidirectional Buck–Boost converter, and an auxiliary storage branch. Unlike conventional off-board bidirectional chargers, the proposed architecture introduces an additional controllable DC-side energy path and treats the EV-side Buck–Boost stage as the main battery-side regulating element. A control-oriented formulation is developed with emphasis on buck/boost operation, current sharing, ripple attenuation, and DC-link coupling, followed by an analytical control framework for the grid-side and DC/DC stages. The system is evaluated under steady-state charging, steady-state discharging, and charging-to-discharging transition conditions. The results confirm stable bidirectional operation, balanced branch-current sharing, and reduced battery-side ripple, with EV-side current ripple of 2.532 A in charging and 2.677 A in discharging. The EV-side converter achieves efficiencies of 90.990% and 89.525%, while the storage-side branch reaches 98.670% and 96.098% in charging and discharging, respectively, and the rectifier efficiency is 94.45% in charging and 90.427% in discharging. In addition, introducing the damping term (R_D) improves grid-current harmonic performance, reducing THD from 4.43% to 1.60% in charging mode and from 0.53% to 0.12% in discharging mode, while preserving IEEE Std 519 compliance. Overall, the proposed architecture provides a technically credible and control-consistent framework for high-power bidirectional EV fast charging.**

***Index Terms*—bidirectional EV fast charging; interleaved Buck–Boost converter; storage-assisted charging system; common DC-link; current sharing; ripple attenuation; harmonic compliance.**

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I. Introduction

The shift to electric transport requires fast, efficient charging infrastructure compatible with grid support. EVs are becoming flexible power system components, aiding renewable energy, bidirectional exchange, and operational flexibility. [1]–[5]. Despite the promise of EVs, wider use is limited by charging infrastructure and technical burdens on power networks [6]–[9]. Bidirectional charging demands high performance in both directions. Off-board fast charging, with better system coordination and robustness, addresses this. Off-board chargers suit medium/high-power EVs by moving power hardware off-vehicle, avoiding on-board limitations and enabling fast DC charging and grid interaction at higher power [10], [11]. Designing fast-charging systems involves more than just choosing converters. Performance relies on the dynamic interaction between AC/DC, DC/DC stages, and the shared DC-link. Good converter design doesn't ensure coordinated system behavior, especially during rapid charging/discharging transitions [12], [13]. This study builds on this observation, but requires more flexible DC-side architectures that better address coupled energy paths than dual-stage chargers.

EV fast-charging research covers power converters, grid interfaces, harmonic reduction, and PI control [12]–[22]. Much

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literature overlooks the coupled interaction of multiple converters sharing a DC-link, focusing instead on individual converter innovation or isolated controller tuning. Even with bidirectional operation, chargers are often seen as conventional two-stage structures with local current control. This view is restrictive when auxiliary storage is added, as the DC-link becomes a shared energy node, not just a buffer. Under these conditions, the EV-side DC/DC converter assumes a broader role in current regulation, ripple shaping, branch balancing, and transient power redistribution.

This architectural and analytical shift provides the immediate motivation for the present work. In our earlier study, a grid-connected bidirectional off-board EV fast-charging system was developed around a three-phase active front-end rectifier, an LCL filter, and a four-phase interleaved bidirectional DC/DC stage, with a unified hierarchical control structure enhanced by feedforward compensation and Artificial Bee Colony (ABC)-based controller tuning [23]. That study established that system-wide coordination is essential for suppressing DC-link instability and improving transient performance, and it demonstrated strong dynamic improvement through optimization of the conventional two-stage off-board charger architecture. In particular, it showed that a unified control treatment can significantly reduce DC-link settling time and maintain very low grid-current distortion under bidirectional operation [23]. However, the earlier work remained centered on a conventional off-board bidirectional charger formulation, in which the overall system was still interpreted primarily as a dual-stage structure. By contrast, the present paper adopts a materially different viewpoint by reformulating the charger as a three-terminal storage-assisted architecture in which an auxiliary storage branch is structurally embedded through the common DC link and in which the EV-side interleaved bidirectional Buck–Boost converter is treated as the principal regulating stage of the battery path rather than as a secondary subsystem.

Accordingly, the main research gap addressed here is not simply the absence of another bidirectional charger topology, but the lack of a structured control-oriented treatment of a storage-assisted fast-charging platform in which the EV-side bidirectional Buck–Boost converter is examined as a central technical element within a shared DC-link system. Once the auxiliary storage path is included, the analytical importance of the EV-side stage is materially elevated: its operating quality influences not only battery-current regulation, but also branch current sharing, effective ripple attenuation, and the transient redistribution of power among the grid-side converter, the EV battery, and the storage branch. Therefore, there remains a clear need for a study that explicitly connects converter-level behavior at the EV side to system-level DC-link dynamics and grid-side performance within a three-terminal fast-charging framework. The present work addresses this need through a control-oriented formulation that focuses on the EV-side interleaved bidirectional Buck–Boost stage in both buck and boost operation, while still preserving a system-level interpretation of the complete architecture.

In light of this gap, this paper presents a three-terminal storage-assisted bidirectional EV fast-charging system coordinated through a common DC link and composed of a grid-side three-phase two-level active rectifier, an EV-side three-branch interleaved bidirectional Buck–Boost converter, and an auxiliary storage-side Buck–Boost interface. Unlike the earlier charger formulation, the technical emphasis is placed on the EV-side converter as the main battery-side regulating stage. The study develops a compact control-oriented model for the system, formulates the EV-side stage in both buck and boost modes, analyzes current sharing and battery-side ripple behavior, and examines the coupling of the converter with the DC-link dynamics. An analytical PI-based control framework is then established for the grid-side and DC/DC stages, and the overall system is evaluated under steady-state charging, steady-state discharging, and charging-to-discharging transition conditions. In this way, the paper aims to demonstrate that a carefully designed analytical control structure can provide stable and technically meaningful performance for a storage-assisted fast-charging architecture while clarifying the operational role of the Buck–Boost-centered EV-side design.

The principal contribution of this work therefore lies in recasting the fast-charging platform from a conventional two-stage off-board charger into a coordinated three-terminal storage-assisted system with an explicitly Buck–Boost-centered interpretation of the EV-side interface. More specifically, the paper:

- proposes a three-terminal storage-assisted bidirectional EV fast-charging architecture coordinated through a common DC link;
- establishes a control-oriented formulation with explicit emphasis on buck/boost operation, branch-current relationships, ripple behavior, and DC-link coupling;
- develops an analytical controller-design framework for the grid-side and EV-side stages; and
- evaluates the architecture under charging, discharging, and mode-transition conditions, with particular attention to current sharing, ripple attenuation, DC-link behavior, and harmonic performance.

Through this integrated treatment, the study extends the design logic beyond the prior off-board bidirectional charger of [23] toward a more flexible and structurally richer storage-assisted fast-charging architecture.

The remainder of this paper is organized as follows. Section II describes the proposed three-terminal storage-assisted fast-charging architecture. Section III presents the control-oriented modeling and control framework. Section IV outlines the simulation setup and evaluation criteria. Section V discusses the steady-state performance, transient response, harmonic behavior, and IEEE Std 519 compliance. Finally, Section VI concludes the paper and highlights future research directions.

II. Proposed Three-Terminal Storage-Assisted Fast-Charging Architecture

This section presents the proposed three-terminal storage-assisted bidirectional EV fast-charging architecture and out-

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lines its principal operating structure. The system is organized around a common DC-link that coordinates energy exchange among the grid-side converter, the EV-side interleaved bidirectional Buck–Boost stage, and an auxiliary storage branch. The section first describes the system configuration and power-flow paths, then highlights the central function of the EV-side interleaved Buck–Boost converter.

A. System Configuration and Power-Flow Paths

The proposed system is structured as a three-terminal storage-assisted bidirectional EV fast-charging architecture established around a common DC-link. As illustrated in Fig. 1, the overall configuration integrates three coordinated power-processing paths, namely a grid-side three-phase two-level active rectifier, an EV-side interleaved bidirectional converter, and an auxiliary storage branch connected through a dedicated Buck–Boost interface. In contrast to conventional bidirectional charger arrangements that are essentially organized as two-port conversion structures [23], the present design introduces an additional controllable DC-side energy path and thereby extends the charger into a more flexible three-terminal platform.

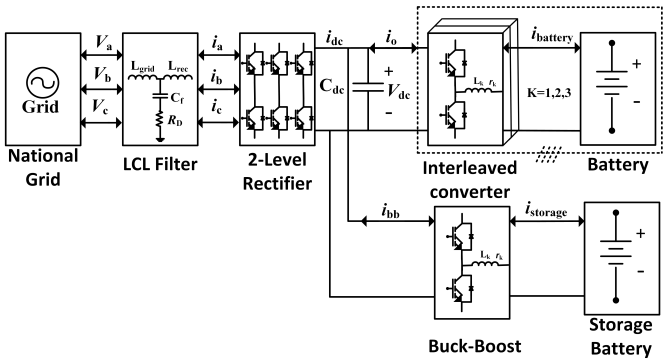


Fig. 1: Integrated architecture of the proposed bidirectional EV fast-charging system with coordinated battery charging and auxiliary energy storage support through interleaved and buck–boost conversion stages.

From a system-level standpoint, the common DC-link serves as the central energy coordination among the utility grid, the EV battery, and the auxiliary storage battery. During charging operation, electrical power is drawn from the three-phase grid, processed by the active rectifier, and transferred to the DC link, from which the EV-side converter regulates the battery charging current. At the same time, the storage branch may absorb or exchange energy according to the intended operating condition. During discharging operation, the direction of power flow at the EV side is reversed, such that the battery-side converter transfers energy from the EV battery back to the DC link, while the grid-side converter operates in the reverse direction to inject power into the utility network.

The electrical realization of the proposed architecture is shown in Fig. 2. The grid-side stage is interfaced with the utility through an LCL filter and is responsible for regulating the DC-link voltage, whereas the EV-side branch employs a

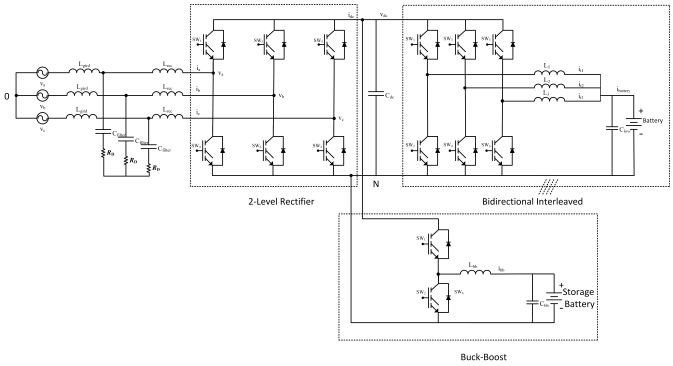


Fig. 2: Electrical configuration of the proposed grid-connected bidirectional EV fast-charging system integrating a three-phase two-level active rectifier, a multi-phase interleaved battery-side converter, and an auxiliary buck–boost energy storage interface.

three-branch interleaved bidirectional Buck–Boost converter to control the EV battery current. In parallel, the auxiliary storage battery is coupled to the same DC link through an additional Buck–Boost stage operating with an independent current reference. The principal system parameters and operating conditions are summarized in Table I.The storage branch is connected to a lower-voltage auxiliary battery and provides an additional controllable power path within the common DC-link framework.

TABLE I: System parameters and operating conditions.

Category	Parameter	Symbol	Value	Unit
Grid-side system	Grid line-to-line voltage	(V_{LL})	415	V
	Grid frequency	(f_g)	50	Hz
	DC-link reference voltage	(V_{dc}^*)	700	V
	Grid-side filter inductance	(L_{grid})	0.133	mH
	Rectifier-side filter inductance	(L_{rec})	0.1397	mH
	Filter capacitance	(C_f)	0.3602	mF
	Damping resistance	(R_D)	0.1	Ω
	DC-link capacitance	(C_{dc})	100	mF
	Switching frequency	(f_{sw})	10	kHz
EV-side Buck–Boost stage	EV battery nominal voltage	(V_{EV})	415	V
	EV branch reference current	(I_{EV}^*)	75	A
	Number of Buck–Boost branches	(N)	3	—
	Branch inductance	(L_k)	1.5	mH
	Branch capacitance	(C_b)	130	mF
	Branch resistance / ESR	(R_k)	2.5	Ω
	Interleaving Phase Shift	Φ	120°	
Storage branch	Storage battery nominal voltage	(V_{st})	206	V
	Storage-branch inductance	(L_{bb})	2	mH
	Storage-branch capacitance	(C_{bb})	1.2	mF
	Storage current reference	(I_{bb}^*)	87.5	A
Operating condition	Aggregated EV-side charging power level	(P_{ch})	135	kW
	Aggregated EV-side discharging power level	(P_{dis})	124.5	kW
	Simulation duration	(T_{sim})	5	s
	Controller sampling time	(T_s)	10^{-6}	s
	Modetransition(G2V→V2G)instant	T	2.5	s

The charging- and discharging-mode power levels reported in Table I represent the aggregated EV-side battery power considering four equivalent battery-side paths. By contrast, the battery power reported in Table III corresponds to the measured power of one EV battery-side path, obtained from the average battery voltage and current.

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B. Central Role of the EV-Side Interleaved Buck–Boost Converter

In storage-assisted fast-charging systems, the technical significance of the battery-side conversion stage extends beyond simple voltage adaptation. Its structure and control quality directly influence charging-current regulation, power-flow reversibility, current ripple, and the manner in which the EV battery interacts with the shared DC-side energy buffer.

Within the proposed system, the EV-side interleaved bidirectional Buck–Boost converter constitutes the principal battery-side power-processing stage. As illustrated in Fig. 2, this converter forms the direct electrical interface between the common DC link and the EV battery pack through three parallel conversion branches. It regulates the battery current, supports branch-current sharing, reduces battery-side ripple, and manages the interaction between the EV battery and the common DC-link, thereby extending its role beyond simple voltage adaptation.

The converter is realized by three bidirectional switching legs, denoted by (SW_1)–$(\mathrm{SW6})$, three branch inductors $((L_1, L_2, L_3))$ with their associated parasitic resistances $((r_1, r_2, r_3))$, a battery-side capacitor $(C\mathrm{low})$, and the EV battery connected at the low-voltage side. This arrangement enables the total battery current to be processed through three parallel energy-transfer paths rather than through a single conversion branch.

III. Modeling and Control-Oriented Formulation

This section develops the control-oriented mathematical formulation of the proposed three-terminal storage-assisted fast-charging system, with particular emphasis on the EV-side interleaved bidirectional Buck–Boost stage. Although the overall architecture comprises a grid-side active rectifier, a common DC-link, an EV-side interleaved conversion stage, and an auxiliary storage branch, the analytical focus is directed toward the dynamic relationships that govern battery-current regulation, branch-current sharing, ripple behavior, and DC-link-coupled energy exchange.

A. Grid-Side Converter Model

The grid-side converter connects to the utility via an LCL filter. This model focuses on grid-current regulation, active-power exchange, and DC-link voltage support. Therefore, the LCL filter is simplified to an equivalent inductive-resistive representation for controller design.

Accordingly, the equivalent grid-side parameters may be expressed as

$$L_s = L_{grid} + L_{rec}, \tag{1}$$

$$R_s = R_{grid} + R_{rec}, \tag{2}$$

where (L_{grid}) and (L_{rec}) the grid and rectifier-side filter inductances, respectively, while (R_s) series resistance.the phase-voltage relations can be described (abc) frame by as the following.

$$v_a = L_s \frac{di_a}{dt} + R_s i_a + v_{aN}, \tag{3}$$

$$v_b = L_s \frac{di_b}{dt} + R_s i_b + v_{bN}, \tag{4}$$

$$v_c = L_s \frac{di_c}{dt} + R_s i_c + v_{cN}, \tag{5}$$

where (v_a), (v_b), and (v_c) are the grid phase voltages, (i_a), (i_b), and (i_c) are the corresponding grid currents, and (v_{aN}), (v_{bN}), and (v_{cN}) denote the converter output voltages referenced to the converter neutral.

For control development, it is more convenient to transform the model into the synchronous (dq) reference frame.

$$v_d = L_s \frac{di_d}{dt} + R_s i_d - \omega L_s i_q + v_{dN}, \tag{6}$$

$$v_q = L_s \frac{di_q}{dt} + R_s i_q + \omega L_s i_d + v_{qN}, \tag{7}$$

The terms $(\pm\omega L_s i_{q,d})$ represent the coupling introduced by the rotating reference frame and therefore appear explicitly in the model. From a control standpoint, these terms are compensated within the current-regulation loop in order to achieve approximately decoupled (d)- and (q)-axis dynamics. With the (d)-axis aligned to the grid-voltage vector, $(v_q \approx 0)$, active and reactive powers in the synchronous frame are written as

$$P \approx \frac{3}{2} v_d i_d, \tag{8}$$

$$Q \approx -\frac{3}{2} v_d i_q. \tag{9}$$

B. Buck–Boost Modeling in Buck and Boost Modes

The EV-side interleaved bidirectional Buck–Boost converter, since this stage governs both the charging and discharging interaction between the EV battery and the common DC link, its mathematical formulation must explicitly reflect the reversal of power flow while remaining sufficiently compact for control design.

As shown in Fig. 1, the EV-side converter consists of three interleaved branches connected between the common DC link and the EV battery. Each branch $(k \in \{1, 2, 3\})$ comprises a bidirectional switching leg, an inductor (L_k), and an associated parasitic resistance (r_k). The three branches operate with a fixed interleaving phase shift and share a common battery-side terminal. Let $(d_k \in [0, 1])$ denote the duty ratio of branch (k), (i_{Lk}) the corresponding inductor current, (v_{bat}) the EV battery terminal voltage, and (V_{dc}) the common DC-link voltage. Under this formulation, the total battery current is obtained from the aggregate contribution of the three branch currents together with the battery-side capacitor current, such that

$$i_{bat} = \sum_{k=1}^{3} i_{Lk} - i_{Cb}, \tag{10}$$

where (i_{Cb}) is the current of the battery-side capacitor. Under steady-state conditions, the average capacitor current is zero, and the battery current is therefore equal to the sum of the branch inductor currents.

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During charging operation, the converter in buck mode, the inductor dynamics of branch (k) may be written as

$$L_k \frac{di_{Lk}}{dt} = d_k V_{dc} - v_{bat} - r_k i_{Lk}, \qquad k = 1, 2, 3, \tag{11}$$

while the battery-side capacitor dynamics are expressed as

$$C_b \frac{dv_{bat}}{dt} = \sum_{k=1}^{3} i_{Lk} - i_{bat}. \tag{12}$$

For control-oriented analysis, linearization around the buck-mode operating point yields the duty-to-current transfer relation

$$G_{id,k}^{buck}(s) = \frac{\hat{i}_{Lk}(s)}{\hat{d}_k(s)} = \frac{V_{dc,0}}{L_k s + r_k}, \tag{13}$$

where $(V_{dc,0})$ is the nominal DC-link voltage at the selected operating point.

During discharging operation, the direction of power flow is reversed and the converter acts in boost mode, transferring energy from the EV battery back to the common DC link, the inductor dynamics of branch (k) become

$$L_k \frac{di_{Lk}}{dt} = d_k v_{bat} - V_{dc} - r_k i_{Lk}, \qquad k = 1, 2, 3, \tag{14}$$

whereas the battery-side capacitor dynamics retain the same form.

In this operating mode, the duty ratio governs the transfer of battery-side energy to the higher-voltage DC-link.

Linearization around the boost-mode operating point gives the corresponding approximate duty-to-current transfer function

$$G_{id,k}^{boost}(s) = \frac{\hat{i}_{Lk}(s)}{\hat{d}_k(s)} = \frac{V_{bat,0}}{L_k s + r_k}, \tag{15}$$

where $(V_{bat,0})$ is the nominal battery voltage at the selected operating point.as in buck operation, the branch dynamics are scaled by the DC-link voltage, whereas in boost operation they are scaled by the battery voltage.

C. Current Sharing, Ripple Behavior, and DC-Link Coupling

In the proposed EV-side conversion stage, the three branch currents are expected to share the total battery current approximately equally. The mean inductor currents may therefore be expressed as

$$i_{L1,\mathrm{mean}} \approx i_{L2,\mathrm{mean}} \approx i_{L3,\mathrm{mean}} \approx \frac{i_{bat,\mathrm{mean}}}{3}. \tag{16}$$

This relation defines the current-sharing objective of the interleaved converter.

Interleaving reduces battery-side ripple by phase-shifting branch currents, causing ripple cancellation at the battery terminal. This lower ripple reduces battery stress, a key advantage of interleaving.

Beyond current sharing and ripple attenuation, the EV-side converter is dynamically coupled to the rest of the system through the common DC-link capacitor. The corresponding DC-link power balance may be written as

$$\frac{d}{dt}\left(\frac{1}{2} C_{dc} V_{dc}^2\right) = P_{ac} - P_{ev} - P_{st}, \tag{17}$$

where (P_{ac}) is the instantaneous power delivered by the grid-side converter, (P_{ev}) is the power processed by the EV-side Buck–Boost stage, and (P_{st}) is the power exchanged with the auxiliary storage branch. Rearrangement yields

$$C_{dc} \frac{dV_{dc}}{dt} = \frac{P_{ac} - P_{ev} - P_{st}}{V_{dc}}. \tag{18}$$

These relations show that the DC-link voltage is governed directly by the instantaneous mismatch among the three power-processing paths.

D. Control Structure and Parameter Determination

The system architecture, consisting of a grid-side active rectifier, an EV-side interleaved bidirectional Buck-Boost converter, and an auxiliary storage branch sharing a DC link, utilizes a hierarchical control strategy. This strategy manages AC-side current shaping and DC-link support at the grid-side, and regulates branch-level current using phase-shifted interleaved operation at the EV-side.

The grid-side control uses a synchronous (dq) reference frame with a PLL for abc-to-dq transformation. The d-axis current regulates active power and DC-link voltage, while the q-axis current is zeroed for near-unity power factor. This cascaded structure uses an outer DC-link voltage regulator to generate the active-current reference, with an inner loop for fast grid current tracking, as shown in Fig. 3.

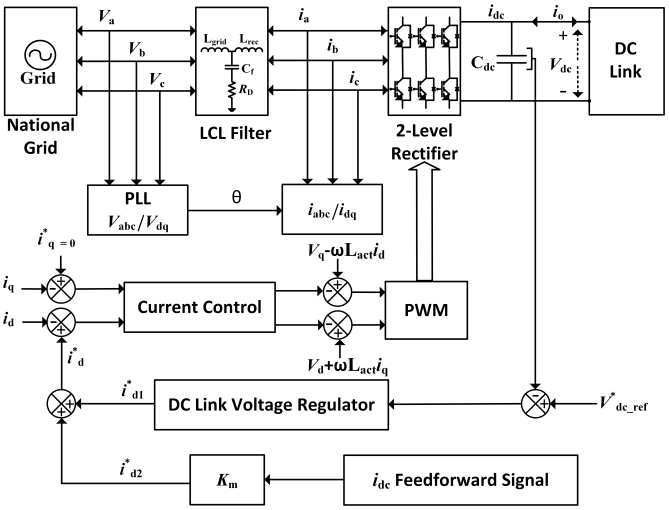


Fig. 3: Grid-side control structure of the two-level active rectifier.

An additional feature of the grid-side controller is the feedforward path introduced through the gain (K_m), as indicated in Fig. 3. The role of this term is to improve transient coordination during rapid power-flow changes, particularly under charging-to-discharging reversal.

The coefficient $K_m = 1.38$ is used as a feedforward scaling factor in the DC-link regulation path. It was selected through time-domain tuning around the rated reference condition of $V_{dc}^* = 700$ V and $I_{dc}^* = 300$ A. The tuning objective was to reduce DC-link voltage excursions during power-flow

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transitions and to support smoother coordination between the grid-side rectifier and the two DC-side Buck–Boost interfaces. The feedforward term therefore complements the outer DC-link PI controller by providing anticipatory compensation for power-demand variations, while the feedback loop remains responsible for eliminating steady-state voltage error.

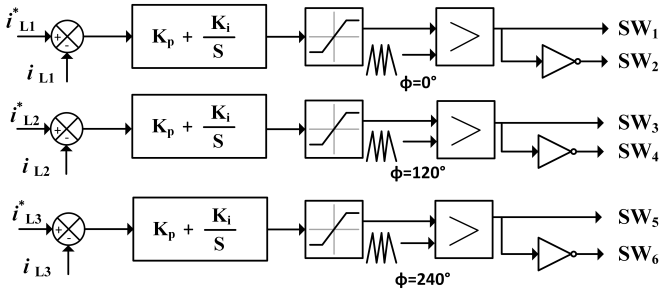


Fig. 4: Per-phase control scheme of the interleaved bidirectional DC–DC converter.

The EV-side interleaved Buck–Boost converter is controlled through a branch-oriented current-control structure, as shown in Fig. 4. Each interleaved branch is assigned an individual PI regulator acting on its inductor current. The branch-current error is defined as

$$e_k(t) = i^*_{Lk}(t) - i_{Lk}(t), \qquad k = 1, 2, 3, \tag{19}$$

and the corresponding PI controller output is given by

$$u_k(t) = K_{p,bi} e_k(t) + K_{i,bi} \int e_k(t)\, dt. \tag{20}$$

This branch-level control formulation is central to the proposed EV-side stage because it simultaneously supports battery-current regulation, branch-current balance, and improved ripple behavior.

To exploit the advantages of interleaving, the PWM carriers of the three branches are shifted uniformly by (120°). The selected carrier phases are therefore

$$\phi_1 = 0^\circ, \qquad \phi_2 = 120^\circ, \qquad \phi_3 = 240^\circ. \tag{21}$$

This phase-shifted modulation causes partial cancellation of the ripple components when the branch currents are summed at the battery terminal.

The controller gains employed in the present study are summarized in Table II.

The implemented control hierarchy follows a time-scale separation principle: the grid-current and branch-current loops provide fast local regulation, whereas the outer DC-link voltage loop performs slower energy-balance correction. In the present conference version, closed-loop behavior is evaluated primarily through full switching time-domain indicators, including DC-link overshoot, rise time, settling time, current ripple, voltage ripple, grid-current THD, and charging-to-discharging transition behavior. A detailed frequency-domain stability-margin study, including loop bandwidths, gain/phase margins, and eigenvalue- or Nyquist-based interaction analysis, is identified as an important extension for future work and for hardware-oriented validation.

TABLE II: Controller parameters.

Control Loop	Parameter	Value
DC-Link Voltage Control	K_p	80.4014
DC-Link Voltage Control	K_i	717.2595
Grid Current Control (d-axis)	K_p	841.2527
Grid Current Control (d-axis)	K_i	130.4289
PLL Synchronization	K_p	0.524
PLL Synchronization	K_i	46.6
DC/DC PI gains	$K_{p,bi}$	0.00785
DC/DC PI gains	$K_{i,bi}$	12.26
Feed forward	K_m	1.38
DC/DC PI gains	$K_{p,bb}$	0.005
DC/DC PI gains	$K_{i,bb}$	10

IV. Simulation Setup and Evaluation Criteria

The effectiveness of the proposed three-terminal storage-assisted fast-charging architecture is evaluated through a targeted simulation framework that reflects key bidirectional EV charging modes and DC-link interactions. The system is assessed under steady-state charging, discharging, and a transition at t=2.5s to examine control reversibility and coordination among the grid-side converter, EV-side interleaved bidirectional Buck–Boost stage, and storage branch. Simulations use predefined conditions (415 V, 50 Hz grid; 700 V DC link). Performance metrics include voltage/current regulation, ripple, DC-link dynamics, grid synchronization, and THD to evaluate power quality, transient response, and overall bidirectional operation

The battery interfaces were represented using generic lithium-ion battery models. The EV-side battery path was parameterized with a nominal voltage of 415 V, rated capacity of 230 Ah, initial SOC of 50%, and internal resistance of $0.018043\ \Omega$. The auxiliary storage battery was parameterized with a nominal voltage of 206 V, rated capacity of 200 Ah, initial SOC of 50%, and internal resistance of $0.0103\ \Omega$. The OCV–SOC-related discharge characteristic was generated internally from the nominal battery parameters rather than from a user-defined lookup table. Temperature and aging effects were not enabled, and the battery response time was set to 0 s; consequently, the simulation emphasizes short-term converter-level electrical and control dynamics rather than electrochemical, thermal, or aging behavior.

Both battery interfaces were operated under a current-regulated constant-current (CC) profile. The EV-side interleaved Buck–Boost converter used a battery-current reference of $I^*_{EV} = 75$ A, while the storage-side Buck–Boost branch used a current reference of $I^*_{bb} = 87.5$ A. A full CC/CV charging transition was not implemented in the present study because the focus is on short-term current regulation, branch-current sharing, ripple attenuation, and common DC-link interaction. SOC-dependent CC/CV operation would modify the

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current reference and voltage constraints over the charging trajectory; therefore, this extension is reserved for future work over wider SOC ranges.

The conversion efficiencies reported in Tables III–V were evaluated consistently from the simulated stage-level power quantities under the corresponding steady-state operating condition. For bidirectional operation, the efficiency values are interpreted according to the relevant direction of power flow in charging and discharging modes. The reported values are therefore presented as stage-level conversion metrics for the rectifier, EV-side Buck–Boost stage, and storage-side Buck–Boost branch.

V. Results and Discussion

A. Steady-State EV-Side and Storage-Side Performance

Steady-state analysis goes beyond voltage and current regulation; it clarifies how effectively each conversion stage sustains controlled power transfer under charging and discharging operation.On the EV side, the result summarized in Table III indicates accurate bidirectional regulation around the intended operating point. These results show that the EV-side interleaved Buck–Boost stage preserves balanced and well-regulated steady-state operation in both directions of power flow while sustaining acceptable conversion effectiveness under the heavier processing burden of the main battery interface.

TABLE III: EV-side Buck–Boost stage battery-side steady-state performance metrics.

Performance Metric	Charging	Discharging
Battery Mean Voltage, $V_{bat,\mathrm{mean}}$ (V)	450.2	443.6
Battery Voltage Ripple, $V_{bat,pp}$ (V)	0.1067	0.120
Battery Mean Current, $i_{bat,\mathrm{mean}}$ (A)	75.01	-74.99
Battery Current Ripple, $i_{bat,pp}$ (A)	2.532	2.677
Inductor Mean Current per Phase, $i_{k,\mathrm{mean}}$ (A)	25.37	-24.54
Inductor Current Ripple, $i_{k,pp}$ (A)	9.007	11.48
Battery Power, P_{bat} (KW)	33.769	-33.266
Conversion Efficiency (%)	90.990	89.525

TABLE IV: Storage-side Buck–Boost stage battery-side steady-state performance metrics.

Performance Metric	Charging	Discharging
Battery Mean Voltage, $V_{bat,\mathrm{mean}}$ (V)	223.9	219.7
Battery Voltage Ripple, $V_{bat,pp}$ (V)	0.9211	0.9684
Battery Mean Current, $i_{bat,\mathrm{mean}}$ (A)	86.88	86.89
Battery Current Ripple, $i_{bat,pp}$ (A)	4.121	8.148
Battery Power, P_{bat} (kW)	19.45	19.09
Conversion Efficiency (%)	98.670	96.098

The storage-side results, summarized in Table IV, indicate similarly stable auxiliary-branch operation, preserves high steady-state conversion effectiveness while providing an additional controllable energy path within the common DC-link structure.

Steady-state results show the EV-side converter handles high-power battery interfacing, minimizing voltage ripple and exhibiting current sharing. The storage-side acts as an energy buffer, showing stable behavior. Rectifier efficiencies 94.45% in charging, 90.427% in discharging, indicate effective grid-side energy transfer, supporting the architecture's practicality.

B. Current Sharing and Ripple Attenuation of the EV-Side Interleaved Converter

Current sharing and ripple attenuation constitute two of the most meaningful steady-state indicators of the effectiveness of the EV-side stage.

The steady-state branch-current waveforms shown in Fig. 5 and Fig. 6 confirm that the three interleaved phases operate in a balanced and well-coordinated manner under both charging and discharging conditions. In charging mode, the three inductor currents exhibit closely matched average values and comparable ripple envelopes, indicating that the total battery current is shared almost uniformly among the parallel branches. A similar behavior is preserved in discharging mode, where the branch currents remain comparably distributed despite the reversal of power flow. This observation is fully consistent with the quantitative values reported in Table III, where the mean inductor current per phase is 25.37 A during charging and (-24.54) A during discharging.

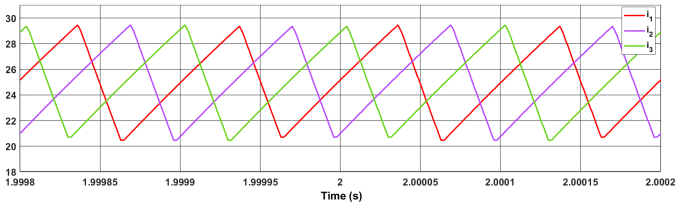


Fig. 5: Steady-state branch currents of the three-phase bidirectional Buck–Boost converter in charging mode.

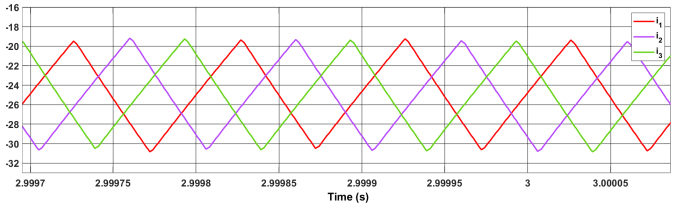


Fig. 6: Steady-state branch currents of the three-phase bidirectional Buck–Boost converter in discharging mode.

The ripple results of the interleaved arrangement. As shown in Fig. 7 and Fig. 8, the battery current remains tightly regulated around its operating reference in both directions of power flow, with only a limited ripple superimposed on the mean current. Table III shows that the battery-current ripple is 2.532 A in charging mode and 2.677 A in discharging mode, whereas the corresponding inductor current ripple per phase is substantially higher, reaching 9.007 A and 11.48 A, respectively. This distinction is analytically significant because it confirms that each branch carries its own switching ripple, yet the phase-shifted summation of the three branch currents yields a substantially smoother aggregate battery current.

The combined evidence of Fig. 5–Fig. 8 and Table III therefore demonstrates that the proposed EV-side stage fulfills its intended role as the core battery-side regulating element of the three-terminal storage-assisted charging system.

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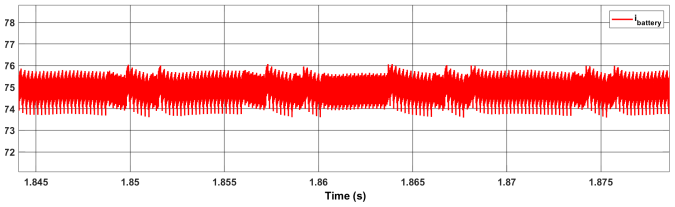


Fig. 7: Battery current ripple in charging mode for the proposed three-phase Buck–Boost converter.

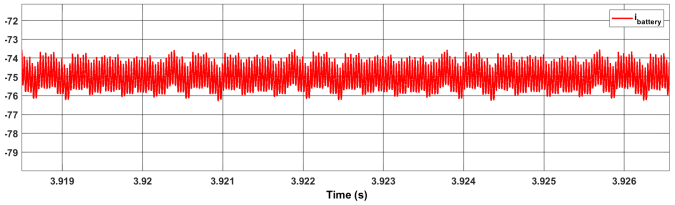


Fig. 8: Battery current ripple in discharging mode for the proposed three-phase Buck–Boost converter.

C. Dynamic Transition and DC-Link Response

The transient response of the EV-side interleaved Buck–Boost converter and the common DC link provides a particularly meaningful measure of the effectiveness of the adopted control framework.

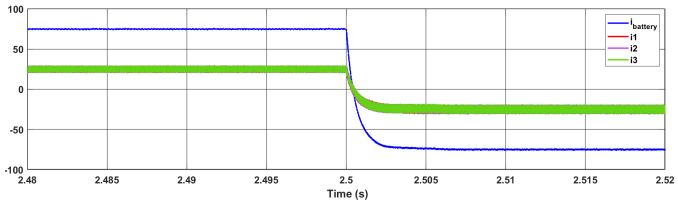


Fig. 9: Dynamic response of the total battery current and individual branch currents during the charging-to-discharging transition.

The dynamic current response shown in Fig. 9 indicates that the total battery current and the three branch currents undergo a smooth and well-coordinated transition at the imposed operating-mode reversal. Prior to the transition instant, the total battery current is maintained at its charging level while the three branch currents remain closely aligned with the expected balanced distribution. Immediately after the transition, the battery current reverses direction and settles to the discharging operating point without evidence of sustained oscillation or pronounced branch divergence.

The corresponding DC-link behavior, shown in Fig. 10, further confirms the stability of the proposed control during mode reversal. At the transition instant, the DC-link voltage experiences a transient excursion, after which it returns toward its nominal operating level in a controlled manner. This behavior is consistent with the expected energy redistribution associated with the reversal from charging to discharging operation.

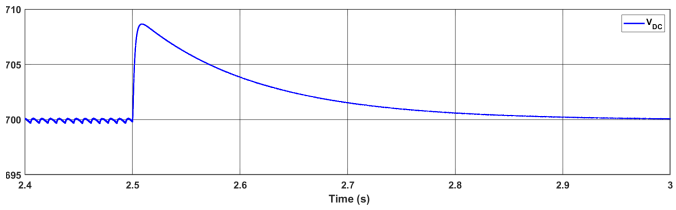


Fig. 10: Dynamic DC-link voltage response during the charging-to-discharging transition.

TABLE V: DC-Link, Current, and Grid-Side Performance.

Category	Parameter	Unit	Charge	Discharge
DC-Link Voltage Performance	Reference DC-link voltage	V	700	700
	DC-link voltage overshoot	V (%)	199.87 (28.55%)	8.7 (1.24%)
	Rise time, t_r	S	0.0125	0.0027
	Settling time, t_s	S	0.2538	0.322039
	Voltage ripple, V_{pp}	V	0.5413	0.06128
	V_{rms}/V_{mean}	V	700	700
DC-Link Current Performance	Mean/RMS DC current	A	300	-300
	Peak overshoot current	A	1000	0
	Rise time, t_r	S	0.00131	0.0007248
	Settling time, t_s	S	0.1790	0.0010193
	Current ripple, i_{pp}	A	9.899	10.87
PLL and Grid-Side Performance	THD total	%	4.43	0.53
	Rectifier conversion efficiency	%	94.45	90.427
	PLL phase deviation	μS	177.759	679.994
	PLL settling time	S	0.0430	0.0232
	Grid current settling time	s	0.2210	0.0142
	Grid current, i_{pp}	A	1092	459.6
	Grid current RMS	A	379.6	156.5
	Grid current mean	μA	769800	8410
	Grid current peak	A	541.2	271.5
	Grid voltage settling time	s	0.037	0
	Grid voltage, V_{pp}	V	1173	1186
	Grid voltage RMS	V	415.5	418.6
	Grid voltage mean	μV	3114	8798

The quantitative results summarized in Table V support this interpretation. The reference DC-link voltage is maintained at 700 V in both operating modes. During charging operation, the DC-link voltage overshoot reaches 199.87 V, corresponding to 28.55%, whereas in discharging operation the overshoot is limited to only 8.7 V, or 1.24%. The associated rise times are 0.0125 s and 0.0027 s, respectively, while the settling times are 0.2538 s in charging mode and 0.322039 s in discharging mode. In addition, the DC-link voltage ripple decreases markedly from 0.5413 V during charging to 0.06128 V during discharging. These values indicate that the proposed system preserves the intended DC-side energy balance under both operating directions.

The DC-link current results further reinforce the same conclusion. The mean DC current changes from 300 A in charging mode to (-300) A in discharging mode, which is fully consistent with the reversal of power-flow direction. The peak overshoot current reaches 1000 A during charging, whereas it is eliminated in discharging operation. Likewise, the current rise time decreases from 0.00131 s to 0.0007248 s, and the settling time decreases from 0.1790 s to 0.0010193 s. The current ripple remains of comparable magnitude, with values of 9.899 A and 10.87 A in charging and discharging operation, respectively. Taken together, these results indicate that the current transition is achieved in a controlled manner and that the reverse operating mode exhibits substantially tighter transient containment at the DC side.

The same table also shows that the transient behavior is closely linked to the grid-side response. The total harmonic distortion is 4.43% in charging mode and 0.53% in discharging

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mode, while the rectifier conversion efficiency is 94.45% and 90.427%, respectively. In addition, the PLL settling time improves from 0.0430 s in charging operation to 0.0232 s in discharging operation, and the grid-current settling time decreases from 0.2210 s to 0.0142 s. The grid current peak also falls from 541.2 A to 271.5 A. These results are analytically important because they show that the charging-to-discharging transition is not resolved only at the battery interface; rather, the entire converter system participates in the transient energy redistribution through the common DC-link.

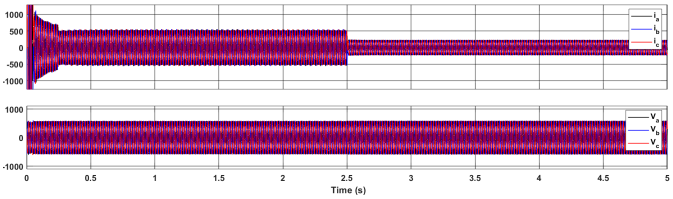


Fig. 11: Overall three-phase grid-side current and voltage response of the proposed storage-assisted EV fast-charging system.

The overall three-phase grid-side response shown in Fig. 11 is consistent with this interpretation. The grid-side current and voltage waveforms remain regular and well behaved over the operating interval.

D. IEEE Std 519 Harmonic Compliance and the Effect of (R_D)

For the power-quality analysis, the damping term (R_D) is examined in terms of harmonic performance and standards-related compliance. This assessment is carried out through the grid-current total harmonic distortion with reference to IEEE Std 519 [24], [25].

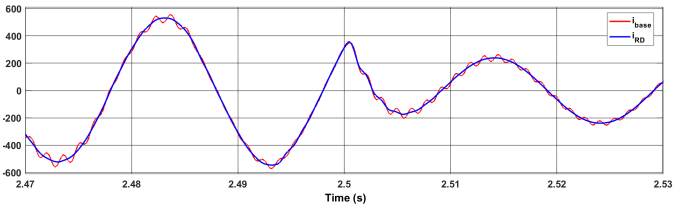
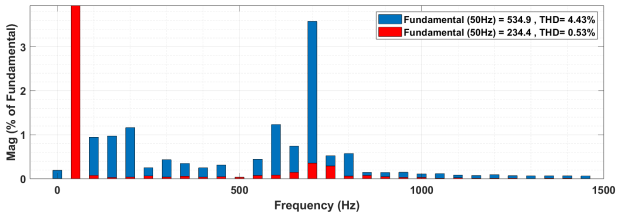


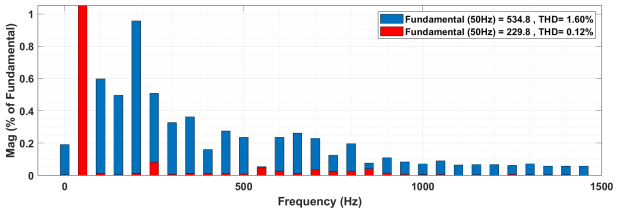
Fig. 12: Comparative grid current response during the charging-to-discharging transition for the baseline and revised control designs.

The comparative current response in Fig. 12 provides an indication of the role of (R_D). During the charging-to-discharging transition, the revised configuration exhibits a smoother and more regular grid-current waveform than the baseline case. The harmonic spectra presented in Fig. 13 show a clearer suppression of non-fundamental components after introducing (R_D).

The quantitative evidence summarized in Table VI confirms this effect more explicitly. In charging mode, the baseline configuration yields a grid-current THD of 4.43%, which already satisfies the adopted IEEE Std 519 reference limit of (5%) [24], [25]. After adding (R_D), the THD decreases further to 1.60%, indicating a substantial enhancement in current quality even though the baseline case is already compliant. In discharging mode, the baseline THD is 0.53%, and the inclusion of (R_D) reduces this value further to 0.12%.



(a)



(b)

Fig. 13: Comparative harmonic spectra of grid current under charging and discharging operation: (a) baseline configuration, (b) after incorporating the damping term R_D.

TABLE VI: Grid-current THD before and after adding R_D with reference to IEEE Std 519 compliance.

Mode	Base (%)	With R_D (%)	IEEE Limit	Base	With R_D
Charging	4.43	1.60	$\leq 5\%$	Yes	Yes
Discharging	0.53	0.12	$\leq 5\%$	Yes	Yes

VI. Conclusion

This paper presents a control-oriented study of a three-terminal storage-assisted bidirectional EV fast-charging system with a common DC link and an EV-side interleaved bidirectional Buck–Boost converter. The system integrates a grid-side three-phase active rectifier, an EV-side interleaved Buck–Boost stage, and a storage-side Buck–Boost branch. The EV-side converter regulates battery current and manages branch-current sharing, ripple attenuation, and DC-link transients.

Steady-state results confirmed controlled DC/DC stage operation. The EV-side converter maintained bidirectional current regulation (75.01 A charging, -74.99 A discharging) with limited ripple (2.532 A, 2.677 A). The interleaved structure balanced branch operation. The auxiliary storage branch provided stable energy support (98.670% efficiency charging, 96.098% discharging). EV-side converter efficiencies were 90.990%and 89.525%, rectifier efficiencies 94.45% and 90.427%. The architecture sustains controlled energy conversion.

Dynamic results showed stability during charging-to-discharging transitions. Battery current reversed controllably, branch currents remained symmetric, and the DC link remained regulated. Charging mode imposed a higher DC-link

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overshoot. The grid-side stage maintained acceptable dynamic behavior with low harmonic content.

Inclusion of a damping term improved grid-current THD (charging: 4.43% to 1.60%, discharging: 0.53% to 0.12%), ensuring IEEE Std 519 compliance. The design supports bidirectional power flow and good grid-current quality.

This work contributes a structured storage-assisted fast-charging framework integrating converter-level behavior and system-level dynamics, coordinating current regulation, ripple shaping, DC-link energy redistribution, and grid-side waveform quality.

Future work should focus on hardware-level validation and experimental verification. Future extensions should also include component-resolved loss mapping over wider load ranges, SOC-dependent CC/CV battery charging profiles, and small-signal loop-interaction analysis using bandwidth, gain/phase-margin, eigenvalue, or Nyquist-based methods. PV-assisted operation and higher-level energy management could further enhance the structure's relevance.

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